MECH MAINT- OFF SITE Department
SIMHADRI

Note No :1

Ref:SMTPP/MECH MAINT-/2023-24/LMI/OFS/1014766 **Date:**20/03/2024(10:38:34)

Subject:LMI Revision-Care and Maintenance of Pumps

Based on OGN-OPS-MECH-026 Rev 0 " The Care and Maintenance of Pumps"

LMI " Care and Maintenance of Pumps Rev 04 " is prepared and submitted for approval.

The schedule of revisions done in Revision 4 is as under may please be approved.

no	Addition in Revision 4	Description of Inclusion	Department
1	Annexure III	Checklist for overhaul of BFP	TMD
2	Annexure IV	Checklist for overhaul of CEP	TMD
3	Annexure V	Checklist for overhaul of CW Pump	OFS
4	Annexure VI	Checklist for overhaul of Sea water Pump	OFS

Submitted by:

टी वी राजशेखर - अपर महाप्रबंधक

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On:20/03/2024(10:38:34) **Ref:**SMTPP/MECH MAINT-/2023-24/LMI/OFS/1014766

Note No:2

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Note No:3

Reviewed and Put up for further approval.

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Note No:4

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On:27/03/2024(10:42:47) **Ref:**SMTPP/MECH MAINT-/2023-24/LMI/OFS/1014766

Note No:5

Clause 4.1 a Resp. to be corrected

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On:27/03/2024(11:30:48) **Ref:**SMTPP/MECH MAINT-/2023-24/LMI/OFS/1014766

Note No:6

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Note No:7

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On:27/03/2024(11:49:31) **Ref:**SMTPP/MECH MAINT-/2023-24/LMI/OFS/1014766

Note No:8

Please review.

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On:27/03/2024(16:44:46) **Ref:**SMTPP/MECH MAINT-/2023-24/LMI/OFS/1014766

Note No:9

Reviewed and put up for approval

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Note No:10

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On:30/03/2024(09:35:37) **Ref:**SMTPP/MECH MAINT-/2023-24/LMI/OFS/1014766

Note No:11

Pl review the LMI and forward for further approval

Submitted by:

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On:30/03/2024(09:52:50) **Ref:**SMTPP/MECH MAINT-/2023-24/LMI/OFS/1014766

Note No:12

Reviewed and put up for urther approval.

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On:30/03/2024(15:54:17) **Ref:**SMTPP/MECH MAINT-/2023-24/LMI/OFS/1014766

Note No:13

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Note No:14

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SIMHADRI

LOCATION MANAGEMENT INSTRUCTION

LMI/OGN/OPS/MECH/026/

ISSUE NO.1; REVISION NO. 04; DATE: FEB'2024

CARE AND MAINTENANCE OF PUMPS

Approved for

Implementation______________________________

HOP (SIMHADRI)

Date: _______________

SIMHADRI

LMI APPROVAL DOCUMENT

TITLE OF LMI	CARE AND MAINTENANCE OF PUMPS
a) DOCUMENT CODE b) REVISION / ISSUE NO & DATE	LMI/OGN/OPS/MECH/026/ ISSUE NO.1, REV.04, FEB'2024
ORIGINAL / REF. DOCUMENT	OGN/OPS/MECH/026
4.PRERPARED BY	DGM (TMD/OFS)
5. CHECKED BY	AGM (TMD/OFS)
6.RECHECKED BY	GM(MAINT)
7. REVIEWED BY	GM(O&M)
8. APPROVED BY	HOP
DATE OF ISSUE	
INDEX LIST	ENCLOSED
DISTRIBUTION LIST	ENCLOSED

INDEX

1.0 INTRODUCTION

Pumps are subjected to a wide range of operating condition with frequent stops & starts. Proper care & maintenance of pumps is vital for the achievement of good unit availability and performance.

This LMI note recommends procedures intended to achieve reliable operation and minimise whole life costs by:

1.1 Establishing and recording the status to provide a basis for assessing their future capability.

1.2 Optimising Operational behaviour and thereby minimising the rate of equipment deterioration and increasing pump efficiency.

1.3 Determining the deterioration in efficiency so that any remedial action required achieving operating targets / efficiency can be implemented on a planned basis at the optimum time.

1.4 Ensuring that routine maintenance and any remedial action employed give the greatest overall benefit.

This document applies to all large centrifugal pumps of turbine area, i.e., Boiler feed pumps, condensate extraction pumps, Sea Water Pumps ,CW pumps.

2.0 REFERENCE DOCUMENT:
COS-ISO-00-OGN/OPS/MECH/026, Rev No: 0 Jan 2024

3.0 SCOPE

This LMI applies to centrifugal pumps like Boiler Feed Pumps, Condensate extraction pumps, Sea Water Pumps, CW pumps.

4.0 Information Required

The Pump status record should contain details of the design intent, manufacture, operational history, maintenance, and modification of each pump.

Resp.: HOD (O) / HOD (TMD/OFS) / HOD (MTP)

4.1 Operational History

The following operational history shall be maintained as per annexure- I & II by the station as mentioned below:

a) Operating hours and number of start ups (hot / worm / cold), condition of recirculation valve passing, ACV position, Steam flow consumption and speed of TDBFPs.

Resp.: HOD (O) / HOD (EEMG).

b) ACV / MCV governing characteristics checking & adjustment.

Resp.: HOD (O) / HOD (TMD).

c) Last cartridge replacement date Resp.: HOD (TMD).

d) Performance test shall be carried out as per requirement of operation/ maintenance Resp.: HOD (EEMG)

e) Vibration data for pre and post overhaul Resp.: HOD (MTP)

f) Details of any abnormal conditions. Resp.: HOD (O) / HOD (TMD/OFS).

g) Problems and failures experienced. Resp.: HOD (O) / HOD (TMD/OFS).

4.2 Inspection and Maintenance

a) Dates of overhauls/ inspections carried out.

Resp.: HOD (TMD/OFS).

b) Summaries of findings in each case including:

- Components inspected and replaced including recirculation control valves.
- Method of inspection e.g. visual, MPI, ultrasonic, mechanical measurement, Clearance etc.,

Resp.: HOD (TMD/OFS).

c) Date of next planned overhaul Resp.: HOD (TMD/OFS).

4.3 MODIFICATIONS:

Details of all modifications and approval taken as per LMI/OD/GEN/006

Resp.: TMD/OFS/MTP

5.0 Plant Operation

5.1 Objective

To identify the parameters requiring limits to minimise component deterioration without compromising plant flexibility.

5.2 General Requirement of all Pumps

5.2.1 Vibration

The routine monitoring of pump vibration can give early warning of impending mechanical Failure. Bearing vibration levels at full load condition should be monitored at monthly intervals for all BFPs & CEPs. The frequency of monitoring should be increased if vibration levels change.

Alarm limits of various categories of pumps are given below: -

MDBFP / TDBFP:

Pump bearing vibration: Alarm> 11 mm / second.

CEP:

Pump vibration: Alarm >9 mm / second

CW pump

Pump vibration:

	Alarm	Trip
Stage 1	80 microns	100 microns (60sec)
Stage 2	180 microns	225 microns (60sec)

Sea Water Pump

Pump vibration:

	Alarm	Trip
Stage 1	09 mm/sec	14 mm/sec (60sec)
Stage 2	09 mm/sec	14 mm/sec (60sec)

5.2.2 Bearing Temperature

MDBFP:

Booster Pump & Main Pump

Journal bearing temperatures: Alarm 90^0C
Trip 95^0C

Thrust bearing temperatures: Alarm 100^0C
Trip 105^0C

Hydraulic coupling

Hydraulic coupling bearing temperature: Alarm 90^0C
Trip 95^0C

Working oil temperature: Alarm 110^0C
Trip 130^0C

TDBFP:

Booster Pump & Main Pump

Journal bearing temperatures: Alarm 90^0C
Trip 95^0C

Thrust bearing temperatures: Alarm 100^0C
Trip 105^0C

CEP

Bearing Temperature : Alarm 100^0C
Trip 105^0C

CW

Thrust Bearing Temperature Alarm 100^0C

Calibration of the instruments for measuring vibration and bearing temperature are done during overhaul.

5.2.3 Net Positive Suction Head

Cavitation is the source of in-service damage to pump internal components and causes a reduction in the pump operational performance both on throughput and efficiency. To limit cavitation damage the Net Positive Suction Head should exceed the NPSH required. The NPSH required should form part of the technical data contained in the record of pump status. The suction head should be monitored and maintained above the minimum level. Also, inlet pressures and temperatures should be monitored while pump in service. The following protections shall be checked during unit overhauling and same shall be recorded for future analysis/audit purposes:

i. Tripping of BFPs on Deaerator level Lo Lo.

ii. Tripping of BFPs on suction valve not open.

iii. Tripping of CEPs oh hot well level Lo Lo.

iv. Tripping of CEPs on suction valve not open.

Resp.: HOD (O)/HOD(C&I)

5.2.4 Over speed / governor Test for Turbine Driven Pumps

Guidance on procedures for operating, testing, and maintaining governors, over speed trips and control valves is given in OD/OPS/SYST/003(Reference 2). The guidance given there in should be followed in the case of boiler feed pump turbines.

Resp.: HOD (O)/HOD(TM)

5.2.5 Pump Casing Distortion

When feed pumps are taken off-load, cooling results in an increase in top to bottom temperature differentials, causing the pump casing to "hog". The Top & bottom body temperature difference should be less than 10 deg C before putting the pump into operation.

Resp.: HOD (O)

6.0 PLANT MAINTENANCE AND MODIFICATION

6.1 Objectives

Maintenance requirement is assessed by monitoring vibration, temperatures, and other performance / efficiency parameters. Accordingly, maintenance is taken up after planning along with MTP. Any modification required for improvement are carried out during overhaul after due approval.

7.0 LIFE ASSESSMENT

7.1 Objectives

To maximise the efficiency of the pump and identify the actions required to assure continued safe, reliable, and efficient operation.

7.2 Condition Monitoring

All the factors mentioned in Section 5.2 should be used to determine the condition of the pump and as part of the life assessment. Additional information should be obtained from pump efficiency test.

7.3 Pump Efficiency

Pump efficiency should be assessed:

- After return from overhaul
- Annually and as per requirement

The result should be recorded and action plan to be made to restore the efficiency.

Resp.: HOD (TMD/OFS)

8.0 EXCEPTION REPORTING

Any exceptions to the instructions specified in the LMI are to be reported as per the following format to AGM (O&M). The exception record shall be maintained by Respective dept.

Resp: HOD (O) / HOD (TMD) /HOD(OFS)/ HOD (MTP)

Description of Exception	Unit No.	Pump name	Remarks

9.0 RECORD KEEPING

The records to be maintained under this LMI include

a) Compliance record of the activities detained under this LMI

b) Exception reporting.

The above records are to be maintained by Turbine maintenance / Offsite HOD.

10.0 REVIEW

Head of the project (HOP) shall review this LMI note once in two years or as and when the reference document is revised.

REVISION INDEX
(FOR CHANGE / REVISION)

SL NO	REV NO	DATE	DETAILS	APPROVED BY
01	**01**	**APR’2013**	Responsibility changed from all DGM to HOD and AGM (O&M) to HOD (O&M)	
02	**02**	**JAN’2020**	**Point no-10:** Review period changed to two yearly basis from three-yearly basis. **Distribution list:** changed	
03	**03**	**FEB’2022**	**S.NO: 4,** In responsibility, HOD (MTP) added **S.NO: 4.1f,** in responsibility, HOD (TMD) added **S.NO: 4.3,** In responsibility, HOD (EEMG) added	
04	**04**	**FEB’2024**	Checklist for BFP/ CEP Pump Overhaul Included. (Annexure-III added)	
05	**04**	**FEB 2024**	Checklist for CW Pumps Overhaul Included (Annexure-V added)	
06	**04**	**FEB 2024**	Checklist for Sea Water Pumps Overhaul Included (Annexure-VI added)	

DISTRIBUTION LIST

SL NO	DATE OF ISSUE	ISSUED TO	COPY NO
		GM (O&M)	
		GM (MAINT)	
		AGM (TMD)	
		AGM (OFS)	
		GM (O)	
		HOD (C&I)	
		HOD (MTP)	
		HOD (EEMG)	

RECORD FOR BFPs AND CEPs
RES: AGM (O) / AGM (EEMG)

SL. NO	DESCRIPTION	TDBFPs (1A,1B,2A,2B, 3A,3B,4A,4B)	MDBFPs (1C,2C,3C,4 C)	CEPs (1A,1B,1C,2A,2 B,2C,3A,3B,3C, 4A,4B,4C)
1	OPERATING HOURS			
2	NO. OF START UPS (HOT/WARM/COLD)			
3	ACV / MCV GOVERNING CHARACTRISTIC CHECKING & ADJUSTMENT			
3	LAST PUMP PROTECTION CHECKING			
4	PUMP EFFICIENCY CHECKING & REPORT			
5	DETAILS OF ANY ABNORMAL CONDITIONS			

RECORD FOR BFPs AND CEPs
RES: AGM (TMD)

SL. NO	DESCRIPTION	TDBFPs (1A,1B,2A,2B, 3A,3B,4A,4B)	MDBFPs (1C,2C,3C,4C)	CEPs (1A,1B,1C,2A,2B,2C,3A,3B,3C, 4A,4B,4C)
1	ACV / MCV GOVERNING CHARACTRISTIC CHECKING & ADJUSTMENT			
2	CARTRIDGE REPLACEMENT/ OVERHAULED DATES			
3	INSPECTION DATES WITH NAME OF COMPONENTS INSPECTED			
4	DATE OF NEXT PLANNED OVERHAUL			

Checklist for Booster/ Boiler Feed Pump Overhaul

Operation	Responsibility	Frequency	Remarks
Inspection of BFP Booster pump internals, weld build-up of casing for any cavitation damage, parting plan matching & replacement of any internals / rotating parts	HOD (TMD)	Overhaul: Every 02 years	
Inspection of BFP Main pump cartridge, replacement of any internals / rotating parts, Dynamic balancing of rotating system	HOD (TMD)	Once in every 10 yrs / EEMG recommendation as per pump efficiency	
Measurement of Journal Bearing clearances	HOD (TMD)	Overhaul: Every 02 years	
DPT & UT of Journal bearings & Thrust pads	HOD (TMD)	Overhaul: Every 02 years	
Check for thrust collar for any scratches/ ensure verticality of collar	HOD (TMD)	Overhaul: Every 02 years	
Measurement of thrust collar run-out	HOD (TMD)	Overhaul: Every 02 years	
Measurement of total axial clearance without thrust bearing and maintain axial position as per design	HOD (TMD)	Overhaul: Every 02 years	
Measurement of total axial clearance with thrust bearing & Adjustment (at rotor axial position)	HOD (TMD)	Overhaul: Every 02 years	
Check oil guard/Catcher clearance	HOD (TMD)	Overhaul: Every 02 years	
Measurement and cleaning of oil inlet line orifices	HOD (TMD)	Overhaul: Every 02 years	
Replacement of gasket of Re-heater spray connection and super heater spray connections	HOD (TMD)	Every 06 year/ Cartridge replacement	
Replacement of gasket of balancing drum line	HOD (TMD)	Overhaul: Every 02 years	
Weld joint DPT of vent line, Seal water line joints, Balancing drum pressure tapping, filter vents, drains etc	HOD (TMD)	Overhaul: Every 02 years	
Proper supporting of all lines specially seal water lines	HOD (TMD)	Overhaul: Every 02 years	
Inspection of Feed water line hangers	HOD (TMD)	Overhaul: Every 02 years	
Visual inspection of Balancing drum lock nut for checking erosion	HOD (TMD)	Every 06 year/ During NDE side mech seal replacement	

Checklist for Booster/ Boiler Feed Pump Overhaul

Operation	Responsibility	Frequency	Remarks
Alignment of pump with Hydraulic coupling/ Drive Turbine	HOD (TMD)	Overhaul/ Opportunity	
Measurement of Distance between coupling faces as per Drg	HOD (TMD)	Overhaul/ Opportunity	
Maintain clearance at holding down bolts	HOD (TMD)	Overhaul/ Opportunity	
Inspection of Barrel seating surface for any cut/erosion etc. Repairing of surface if required.	HOD (TMD)	During Cartridge replacement	
Replacement of all gaskets during every Cartridge replacement	HOD (TMD)	During Cartridge replacement	
Measurement of clearances at all areas as per drawing i.e. wear rings and balancing drum area	HOD (TMD)	During Cartridge replacement	
Replacement of soft packing/O-ring during mech seal replacement.	HOD (TMD)	During mech seal replacement	
Protection and interlock checking	HOD (O) & HOD (C&I)	Overhaul/ Opportunity	
Measurement of Vibration	HOD (MTP)	Monthly	
Ensure tripping (Manual/Protection) of BFP Main pump, Booster Pump, Gear Box etc as per tripping value on high vibration / bearing temperature as recommended by OEM	HOD (O)	Immediately	
Servicing of Feed water outlet valves and suction valve	HOD (TMD)	Every COH	
Check passing of Discharge NRV	HOD (O) & HOD (TMD)	Before every OH	
Inspection of BFP RC Valves	HOD (TMD)	Overhaul/ As per recommendation of EEMG	
Cleaning of both suction strainers	HOD (TMD)	Overhaul/ Opportunity	
Check suction valve limit switches and monitoring in UCB	HOD (O) & HOD (C&I)	Every Overhaul	
BFP Efficiency test to be carried out	HOD (EEMG)	Quarterly	
Checking of BFP RC valve for passing	HOD (O) & HOD (EEMG)	Fortnightly	

Checklist for Condensate Extraction Pump Overhaul

Operation	Responsibility	Frequency	Remarks
Servicing of Complete pump after removing from position, Measurement of control dimensions, replacement of internals such as all soft parts, Cutless rubber bearing, any other damaged parts	HOD (TMD)	As per EEMG recommendation based on specific energy consumption / condition monitoring	
DPT & UT of Journal cum thrust bearings pads	HOD (TMD)	Overhaul: Every 02 years	
Check for thrust collar for any scratches/ ensure verticality of collar	HOD (TMD)	Overhaul: Every 02 years	
Measurement of total axial clearance without thrust bearing and maintain axial position as per design	HOD (TMD)	Overhaul: Every 02 years	
Measurement of total axial clearance with thrust bearing & Adjustment (at rotor axial position)	HOD (TMD)	Overhaul: Every 02 years	
Check oil guard/Catcher clearance	HOD (TMD)	Overhaul: Every 02 years	
Check lub oil cooler for leakage (hydro test)	HOD (TMD)	Overhaul: Every 02 years	
Check Orifice of seal water line	HOD (TMD)	Overhaul: Every 02 years	
Proper supporting of all lines specially seal water lines	HOD (TMD)	Overhaul: Every 02 years	
Alignment of pump with Motor	HOD (TMD)	Overhaul/ Opportunity	
Measurement of Distance between coupling faces as per Drg	HOD (TMD)	Overhaul/ Opportunity	
Replacement of soft packing/Oring during mech seal replacement.	HOD (TMD)	During mech seal replacement & Every OH	
Protection and interlock checking	HOD (O) & HOD (C&I)	Overhaul/ Opportunity	
Ensure tripping (Manual/Protection) of pump on High vibration / bearing temperature as per tripping value recommended by OEM	HOD (O)	Immediately	
DPT of inlet bellows	HOD (TMD)	Overhaul: Every 02 years	
Servicing of Condensate outlet valves	HOD (TMD)	Every COH	
Servicing of Condensate suction valve	HOD (TMD)	Overhaul: Every 02 years	

Check passing of Discharge NRV	HOD (O) & HOD (TMD)	Before every OH	
Checking passing of RC valve an rectification	HOD (O) & HOD (TMD)	Before every OH	
Check suction valve limit switches and monitoring in UCB	HOD (O) & HOD (C&I)	Every Overhaul	
Cleaning of suction strainers	HOD (TMD)	Overhaul/ Opportunity	
Ensure installation of Nozzles in discharge of recirculation line inside condenser	HOD (TMD)	Overhaul/ Opportunity	

Checklist for Volute type CW Pump Overhaul

Activity	Responsibility	Frequency
Dewatering of Pump Suction Bay & Removal of Sludge	HOD(OFS)	yearly/every overhaul
Inspection of discharge pipe by opening manhole for ensuring discharge valve not passing	HOD(OFS)	yearly/every overhaul
Complete Overhauling of CW Pump based on rolling plan	HOD(OFS)	Major 5 years Minor Yearly
PM of remaining CW Pumps to ensure availability	HOD(OFS)	
Complete Servicing of CW Discharge ButterflyValves and its Hydraulic System	HOD(OFS)	yearly/every overhaul
Fitness of Divers to be ensured	HOD(OFS)	yearly/every overhaul
All essential PTW's to be ensured	HOD(OFS)	yearly/every overhaul
Cleaning & Painting of Trash Rack	HOD(OFS)	yearly/every overhaul
Servicing of Lube water pumps/Seal water pumpsand its associated valves	HOD(OFS)	yearly/every overhaul
Servicing of CW Drain Pit pumps	HOD(OFS)	yearly/every overhaul
Ensuring service water/Lube water header healthiness	HOD(OFS)	yearly/every overhaul
Inspection & NDT of Critical Pump Internals	HOD(OFS)	yearly/every overhaul
Surface straightness check on motor stool and discharge head using straight edge & master level	HOD(OFS)	yearly/every overhaul
Blue matching of shaft-couplers, muff, Discharge head-pump sole plate	HOD(OFS)	yearly/every overhaul
Inspection of Pump REJ for stage 2 pumps & change it if required	HOD(OFS)	yearly/every overhaul
Inspection, Repair & coating/painting of CW discharge portion	HOD(OFS)	yearly/every overhaul
Inspection of Filters, cleaning & replacement of filter elements in lube water system & seal water system	HOD(OFS)	yearly/every overhaul
Joint Protocol of CW Discharge Butterfly valve opening, Emergency closing & Normal closingtimings by Offsites, CnI & Operation	HOD(OFS)	yearly/every overhaul

Checklist for Sea Water Pump Overhaul

Activity	Responsibility	Frequency
Decoupling & No load Checking	HOD(OFS)	3-yearly/5 yearly
Dismantling of Motor & it's Connections	HOD(OFS)	3-yearly/5 yearly
Dismantling of C&I connections	HOD(OFS)	3-yearly/5 yearly
Motor removal & Dismantling of pump	HOD(OFS)	3-yearly/5 yearly
Material mobilization to sea water pump house based on requirement	HOD(OFS)	3-yearly/5 yearly
Complete overhaul of Bowl Assembly	HOD(OFS)	3-yearly/5 yearly
Shaft Sleeves and bearing assembly Overhaul	HOD(OFS)	3-yearly/5 yearly
Pump internals cleaning & Painting	HOD(OFS)	3-yearly/5 yearly
Pump Assembly including Motor Stool	HOD(OFS)	3-yearly/5 yearly
Motor placement & Alignment Checking	HOD(OFS)	3-yearly/5 yearly
Inspection & NDT of Critical Pump Internals	HOD(OFS)	3-yearly/5 yearly
Surface straightness check on motor stool and discharge head using straight edge & master level	HOD(OFS)	3-yearly/5 yearly
Blue matching of shaft-couplers, muff, Discharge head-pump sole plate	HOD(OFS)	3-yearly/5 yearly
Inspection, Repair & PU coating/painting of Sea water discharge portion	HOD(OFS)	3-yearly/5 yearly
Inspection , cleaning & replacement in lube water system & dosing system	HOD(OFS)	3-yearly/5 yearly
Checking of Sump for Travelling screen chain and sprocket	HOD(OFS)	3-yearly/5 yearly